Passages - White Holes

There is nothing wrong with changing your mind, given that we are all still learning. The best scientists are the ones who change their minds often, as Einstein did.

We have access only to perspectives. Reality is perhaps nothing other than perspectives. There is no absolute. We are limited, impermanent. and precisely for this reason, to live, to be, as we do, is so light and sweet ...

We are always convinced that our natural intuitions are self-evidently right, and it is this that prevents us from learning more. The difficulty lies not in learning, but in unleaming.

Falling is a direct consequence of the slowing of time. It would take a bit of math to show in detail how, but a stone falls precisely because it follows a straight trajectory in spacetime distorted by a local slowing-down of time.

With the voice of Ulysses in our ear: Do not pass up the chance to experience, beyond the sun, the world where no one lives. Consider your origin: you were not made to live like brutes, but to pursue virtue and knowledge. And like the companions of Ulysses, on this mad flight we'll make wings of our oars.

However enormous it might be, the length of the interior of a black hole is not infinite: at the bottom there is still the star that, by collapsing in on itself, gave rise to the hole.

A singularity is in the future. As the diameter of the funnel narrows, the

cylinder becomes more curved, like a roll that is rolled tighter. The more the funnel is narrowed, the greater the distortion of spacetime. When this reaches the fateful "Planck scale," the scale where we expect space and time to be seriously affected by quantum phenomena, we enter the region where these phenomena imply that Einstein's equations are violated. If we ignore these phenomena and trust in Einstein's theory; the equations predict that the crushing of space continues until catastrophe occurs: the long, thin tube becomes longer and thinner until it is squeezed into a single line (crushing us with it).
And then? Then that's it. Space has collapsed, time is finished. We hit a wall.
If we go along with Einstein's theory, time ends here.

Hipparchus imagines flying to the tip of the cone made by the Earth's shadow, then looking back.
Looked at from there, the Earth hides the Sun perfectly. Hence the angle a is half the angle within which we see the Sun. The angle is half the angle under which we see the Moon. But Sun and Moon
appear to be of equal size in the sky, hence a = B. Euclid's geometry informs us, then, that the two
dotted lines are parallel, and the drawing shows that the radius of the Moon plus the radius of the shadow (where the Moon is) makes a segment equal to the radius of the Earth. Observing an eclipse shows that the radius of the disk of the shadow is two and a half times the radius of the Moon, hence
• the radius of the Earth is three and a half times the radius of the Moon. A coin 1 centimeter in diameter covers the Moon if we hold it 110 centimeters from our eye

(try it for yourself!), hence the distance to the Moon is 110 times its diameter. Hence, the distance to the Moon is 110 divided by 3½
-that is, around thirty times the diameter of the Earth. And this is exactly right! Pure genius. All this from simple observations that we can make in our garden with the naked eye.

How can we go and see from somewhere that we cannot actually reach?
I think that the answer is to grope for a delicate balance. A balance between how much we take with us and how much we leave at home. What we carry with us allows us to know what to expect. These are the maps, the rules, the generalities that we trust in because they have worked so well. And yet, we know that we must leave something behind. If we leave too many things at home, we lack the tools needed to forge ahead; if we take too many, we fail to find the paths to new understanding. There are no recipes for success; there is only trial and error. Trying and trying again. This is what we do . . . the long study and the great love.

Making an analogy involves taking an aspect of a concept and using it in another context, preserving something of its original meaning while letting something else go, in such a way that the resulting combination produces new and effective meaning. This is how the best science works.

A white hole is not even another solution of Einstein's equations. It is the same solution that describes a black hole, reversed in time. The same solution with the sign of the time variable flipped. The same solution, as seen backward in time. A white hole is how a black hole would appear if we could film it and run the film in reverse.

If space is granular, the inside of a black hole cannot be squashed any further than the size of the individual grains. There is a limit to compression and distortion. The contraction that squeezes the tube of the inside of a black hole must therefore stop before the singularity.

Things are not always definite. A particle does not always have a position. Sometimes it can be nowhere, disembodied like a wave, before materializing somewhere else: it can leap. One consequence of this leapy aspect of reality is the phenomenon known as the "tunnel effect." This is the ability that things have to cross barriers that would be insurmountable if it weren't for the quantum.

The quantum properties of space and time allow the inside of the black hole to "leap" beyond the singularity, when classical equations would have time stop. Quantum leaps are well known in physics. Since the early days of the quantum saga, Niels Bohr realized that atoms emit light when electrons "jump" from a larger orbit to a smaller one. But here the quantum leap is far more radical than a particle hopping around from one location to another: here it is spacetime itself that jumps.
The leap of spacetime is not a phenomenon that takes place in space and in time. It is a phenomenon that is neither spatial nor temporal: it is an instantaneous quantum transition from one configuration of space to another.

Gravitational attraction does not become repulsion by reversing the direction of time.

This is the magic of the elasticity of time. The black and the white horizons are themselves different, but from the outside they look precisely the same. horizons and the insides distinguish white from black, future from past; the outside does not.

Heat emission is the most characteristic of the irreversible processes: the processes that occur in one time direction and cannot be reversed. Heat distinguishes past from future.

Hawking radiation is a phenomenon that regards mainly the horizon, not the deep interior of the hole. Therefore, a very old black hole turns out to have a peculiar geometry: an enormous interior (that continues to grow) and a minuscule (because it has evaporated) horizon that encloses it.

On the basis of these arguments, many physicists have accepted a "dogma" the number of elementary components contained in a small surface is necessarily small. Where does the error lie? It lies in the fact that both arguments refer only to the components of the black hole that can be detected from the outside, as long as the black hole remains what it is. And these are only the components residing on the horizon. The description of a black hole from the outside is incomplete.

The irresistible flow of time is a reflection of a way in which things happen to be arranged around us.

If the present has traces of the past, it is due solely to th disequilibrium of that past. It is for this reason that we remember the past and not the future because o the disequilibrium in the past. We know the past because there are traces of it i the present-in our memories, for example. To say that the past is determined i to say no more than that we have many traces of it. It is not a direction intrinsi to time that makes the past knowable, determined: what we call the past is hov things were arranged at one point in time. It is the disequilibrium of the past-only that--that gives rise to traces.

Spinoza writes: "Men feel free, because they are aware of their choices and their wishes, but they ignore the causes that lead them to will and to choose, and do not give the slightest attention to these causes." And again: "Men think that of their own free will they can do something or refrain from doing it, opinions that consist only of this, that . . . they ignore the causes that make them act as they do."

This brings us to a conclusion that seems to me extraordinary. Our neurons, our books, our computers, the DNA in our cells, the historical memory of an institution, the entire contents of the data on the internet, my sweet guide, whose holy eyes were glowing as she smiled, the ultimate source of all the information of which life, culture, civilization are made, is none other than the disequilibrium of the universe in the past.